LLM-Condition / Research-Result Boundary

Execution Conditions, Auditability, and Reproducibility in LLM-Mediated Research

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Abstract

This document formalizes the **LLM-Condition / Research-Result Boundary** as a protocol-facing note for research in which a large language model materially contributes to data production, annotation, simulation, intervention, coding, classification, analysis, or interpretation. The central claim is that an LLM-mediated result cannot be adequately interpreted apart from the conditions under which the model was executed.

A broad model name does not specify a model condition. Provider, version, date of access, interface, system-level instructions, prompt structure, configuration, memory, retrieval, input material, post-processing, validation, and degree of automation can all alter the resulting evidence. When these conditions are absent or unstable, a reported output may remain readable while becoming difficult to audit, compare, reproduce, or delimit methodologically.

The note distinguishes an LLM output from a research result. Output becomes research-relevant only when its role, production conditions, validation status, processing history, and interpretive boundary are made explicit. The framework does not claim that complete computational reproducibility is always possible, especially in changing proprietary systems. It instead defines a minimum structural requirement: the result must carry enough of its execution field for another reader to understand what was done, what could vary, what was checked, and which claims remain bounded.

The note can support reporting protocols, methods sections, preregistration design, replication review, and institutional assessment without functioning as a platform-specific tutorial or a universal reproducibility standard.

Keywords

LLM Research; Model Condition; Research Result Boundary; Reproducibility; Auditability; Prompt Disclosure; System Instructions; Model Version; Access Mode; Human Validation; Automation Degree; Methods Reporting; Execution Field; Meta-Writing Ecology.

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1. Introduction

LLM-mediated research often produces stable-looking artifacts: a classification table, a set of simulated responses, an annotation file, a conversational intervention, a coded corpus, or a textual analysis. The visible artifact can be stored and quoted. The system that produced it may not be equally stable.

Unlike a fixed instrument, an accessed model can change while retaining the same public label. Interfaces can add hidden instructions, memory, moderation, retrieval, or tool conditions. Repeated runs can vary. A web interface and an API can create materially different environments. Human selection and post-processing can further shape what is finally reported.

The methodological object is therefore not the output alone. It is the output together with the execution conditions that generated, filtered, selected, and validated it. This note defines that boundary and translates it into an auditable reporting orientation.

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2. Core Definition

LLM-Condition / Research-Result Boundary describes the condition in which a research result produced with, through, or about a large language model is interpreted without preserving the execution conditions under which that result was generated.

The issue is not that LLMs are used in research. LLMs may be used legitimately as research tools, annotators, simulators, intervention systems, participant-facing agents, stimulus generators, coding assistants, data-processing systems, or empirical objects of study.

The boundary problem appears when the result is reported as if it were separable from the model condition that produced it.

An LLM-based result depends not only on the visible output, but also on a layered execution environment:

model identity model version provider access mode interface condition system prompt user prompt configuration runtime environment memory state input material post-processing human validation automation degree reproducibility materials

When these conditions are missing, vague, unstable, or unreported, the result may remain readable but not reproducible, interpretable but not auditable, and usable but not methodologically bounded.

The central structural claim is:

LLM-mediated evidence must carry its execution conditions.

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3. Classification and Scope

This document presents **LLM-Condition / Research-Result Boundary** as a protocol-facing boundary note. It identifies the conditions a reporting, review, or documentation procedure should keep visible, but it is not itself a complete protocol, checklist standard, certification system, or compliance instrument.

The note can support method design and audit preparation without prescribing a single technical implementation. The public version omits non-public editorial records, private source materials, non-public working materials, and implementation details.

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4. Central Distinctions

model name ≠ model condition

Expanded:

A broad model label does not identify the actual system, version, access mode, configuration, prompt environment, memory state, or runtime condition under which a result was produced.

Second distinction:

LLM output ≠ validated research result

Expanded:

An output generated by an LLM does not become a research result until its role, input condition, validation status, post-processing, reproducibility boundary, and interpretive scope are made explicit.

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5. Core Mechanism and Structural Sequence

5.1 Core Mechanism

research task incorporates LLM → model produces output or mediates procedure → output appears analyzable → model label is reported at a broad level → prompt, configuration, access mode, memory, or validation condition remains partial → result is interpreted as stable → replication or audit fails → research-result status exceeds disclosed model condition

5.2 Structural Sequence

Failure Flow

LLM used in research workflow → output generated → broad model name reported → prompts / system instructions / access conditions incompletely documented → validation remains partial or absent → result enters paper as evidence → readers cannot reconstruct execution condition → output is treated as research result without auditability

Aligned Flow

LLM used in research workflow → role of LLM declared → model condition specified → prompt and system instructions reported → configuration and access mode disclosed → input and privacy handling documented → human validation or interpretation boundary stated → code / scripts / examples shared where possible → result remains tied to execution condition → auditability is preserved

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6. Structural Components

LLM role research task model name provider model version date of access access mode API / web / local interface context mode system prompt user prompt prompt template configuration temperature seed token limit number of runs customization fine-tuning retrieval augmentation persistent memory input data sensitive data data contamination risk post-processing human validation automation degree code sharing script sharing example interaction competing interests reproducibility boundary

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7. Primary Failure Patterns

1. Model-Name Sufficiency Error

   "LLM used" → broad model label reported → execution condition hidden

   A model family name is treated as enough methodological detail.

2. Prompt-Only Reproducibility Error

   prompt reported → reproducibility assumed

   The prompt is disclosed, but system instructions, model version, access mode, runtime environment, memory, or configuration remain hidden.

3. Version Drift

   model updated → same label remains → output behaviour changes

   The model name appears stable while the underlying model condition changes.

4. Interface Drift

   same model → different access mode → different behaviour

   API, web interface, local deployment, chat mode, or separate-call mode may produce different execution conditions.

5. Configuration Blindness

   temperature / token limit / seed / run count omitted → output variability unbounded

The study reports the result but not the conditions affecting output variability.

6. Memory Contamination

prior interaction persists → later output influenced → study condition misreported as independent

Persistent memory or prior context modifies outputs without appearing in the method.

7. Training-Exposure Ambiguity

model may have seen test material → performance interpreted as task competence

Possible exposure to stimuli, benchmarks, or study materials remains unaddressed.

8. Construct Validation Gap

LLM classifies construct → classification accepted → human validation missing or weak

The model output is treated as measuring the intended construct before validation.

9. Automation Responsibility Blur

workflow automated → human role unclear → accountability ambiguous

The degree of automation is reported insufficiently, making responsibility hard to place.

10. Reproducibility Surface Without Reproducibility Depth

parameters reported → reproducibility claimed → model access / version / infrastructure unavailable

Reporting exists but does not restore the conditions needed for replication.

8. Operational Evaluation

1. What role did the LLM play in the research workflow?
2. Was the LLM a tool, participant simulator, intervention, annotator, coder, analyst, chatbot, stimulus generator, or object of study?
3. What exact model, provider, version, and date of access were used?
4. Was the model accessed through an API, web interface, local deployment, or another controlled environment?
5. Were system-level instructions present?
6. Were exact prompts, prompt templates, or interaction structures reported?
7. Were configuration settings reported where available?
8. Did the session include memory, prior context, persistent state, or cross-interaction carryover?
9. Were inputs, preprocessing, sensitive data handling, and privacy constraints documented?
10. Was the LLM output validated by humans or by another independent procedure?
11. Were post-processing steps applied?
12. Were code, scripts, notebooks, or example interactions made available?
13. Could another researcher reconstruct the execution condition?
14. Was the degree of automation disclosed?
15. Who remains responsible for interpretation, validation, and downstream claims?

9. Application Domains

1. LLM annotation and coding

Reported labels require disclosure of model condition, prompt structure, validation procedure, disagreement handling, and human review.

2. Synthetic participants and simulations

The evidential role of generated responses must remain tied to the model, persona, sampling, grounding, and validation conditions.

3. Participant-facing agents

Interventions delivered through an LLM require documentation of system instructions, memory, safety layers, versioning, and session conditions.

4. LLM-assisted analysis

Summaries, thematic analyses, coding assistance, and interpretive transformations require a visible boundary between generated candidate material and validated research judgment.

5. Replication and audit

Even when exact reproduction is impossible, a documented execution field permits meaningful comparison, limitation, and partial reconstruction.

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10. Relations to Adjacent Public Frameworks

1. Generation-Condition Disclosure–Reproducibility Cross

Generation-Condition Disclosure–Reproducibility Cross provides an adjacent public framework for locating the present structure.

2. Provenance–Validity Separation Model

Provenance–Validity Separation Model separates traceable origin from the validity, adequacy, or warranted use of an output. The present entry extends that distinction into a more specific interface, research, or responsibility condition.

3. Verification Labor Compression

Verification Labor Compression describes how accelerated production can move validation work downstream, hide it, or concentrate it at a final review node. The present entry identifies the particular object or boundary that the verification work must preserve.

4. Responsibility Alignment Model

Responsibility Alignment Model clarifies whether ownership, authority, capacity, visibility, and accountability remain attached to compatible nodes. The present entry specifies one condition under which that alignment can fail or become nominal.

5. AI-Readable Knowledge Architecture

AI-Readable Knowledge Architecture concerns the structuring of public knowledge so that machine-mediated reading does not erase provenance, limits, or relation status. The present entry supplies a boundary that such architecture should keep visible.

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11. Boundary Conditions and Non-Applicability

11.1 Boundary Conditions

This note applies when an LLM is materially involved in the production, classification, simulation, intervention, annotation, analysis, coding, summarization, or interpretation of research evidence.

The boundary is crossed when:

LLM output → treated as research result

without sufficient documentation of the conditions under which the output was generated.

It also applies when:

model name → treated as sufficient methodological reporting

or:

prompt disclosure → treated as sufficient reproducibility

The note does not apply to minor editorial or stylistic assistance unless that assistance materially affects research design, evidence production, data interpretation, participant interaction, coding, or analysis.

11.2 Non-Applicability

This note should not be used to reject all LLM-based research, all LLM-assisted workflows, or all studies using commercial systems.

It does not apply strongly when:

LLM use is purely editorial and does not affect evidence

LLM output is not part of data generation, analysis, interpretation, or intervention

the research result does not depend on the LLM condition

the LLM role is trivial, disclosed, and methodologically irrelevant

the work is exploratory and explicitly not presented as reproducible evidence

execution conditions are proportionately documented for the use case

This note should not be reduced to:

LLM research is unreliable

or:

full reproducibility is always possible

The narrower concern is:

LLM-mediated research evidence loses auditability when execution conditions are not preserved.

LLM Reporting Auditability Protocol Anchor

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12. Public Orientation and Method Boundary

This document is designed as a public conceptual entry. It provides a definition, mechanism, diagnostic orientation, and limit statement. It omits non-public editorial records, private source materials, non-public working materials, and implementation details.

The entry is not a substitute for empirical research, legal analysis, professional judgment, technical testing, or domain-specific governance. It should not be used to score people or institutions, automate high-stakes decisions, or infer misconduct from conceptual similarity.

Examples and application domains are illustrative. A case should be classified under **LLM-Condition / Research-Result Boundary** only when the relevant structure can be demonstrated from available evidence.

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13. Limitations

1. Dynamic systems

A documented model condition may later become inaccessible, limiting exact replication.

2. Undisclosed provider layers

Researchers may not have access to all system instructions or infrastructure details.

3. Reporting is not validation

Complete disclosure does not establish that the construct, inference, or conclusion is valid.

4. Privacy and security constraints

Prompts, inputs, or system details may require redaction or controlled access.

5. No single reporting minimum

The necessary level of detail varies with the model's methodological role and the consequence of the claim.

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14. Minimal Formulation

model name ≠ model condition

same prompt ≠ same execution condition

prompt text ≠ full instruction environment

LLM output ≠ validated construct

reported result ≠ reproducible procedure

parameter reporting ≠ full reproducibility automation degree ≠ responsibility clarity

model provenance ≠ result validity

configuration disclosure ≠ construct validation

API access ≠ web-interface condition

output similarity ≠ methodological equivalence

system prompt absence ≠ instruction absence

temperature setting ≠ execution stability

memory state ≠ neutral context

LLM-assisted analysis ≠ human-validated analysis

research transparency ≠ interpretive adequacy

the output is not the method

the model condition is part of the evidence

The output is not the method.

Expanded:

An LLM-based research result cannot be interpreted apart from the model, prompt, access, configuration, input, validation, memory, and reproducibility conditions under which it was produced.

Alternative compressed form:

The model condition is part of the evidence.

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15. Conclusion

The output is not the method. In LLM-mediated research, the model condition forms part of the evidence because it shapes what can be generated, selected, interpreted, and reproduced. A methods section that names only a model family can leave the decisive execution field invisible.

The boundary proposed here does not demand impossible transparency. It requires proportionate disclosure of the conditions that materially affect the result, explicit validation of the output's research role, and clear marking of what another researcher can and cannot reconstruct.

Naming Declaration

LLM-Condition / Research-Result Boundary is a Meta-Writing Ecology term for the boundary between an LLM-mediated research result and the execution conditions required to interpret, audit, validate, or reproduce that result.

The term LLM condition refers to the model, version, provider, access mode, interface, configuration, prompt environment, system instructions, memory state, input condition, customization, post-processing, and validation context under which an LLM output is produced.

The term research result refers to any finding, classification, simulation, annotation, intervention effect, generated stimulus, coded construct, analytic output, or empirical claim that depends materially on LLM use.

The term does not refer to any specific model provider, research domain, social science method, behavioural science task, benchmark, platform, or checklist.

Within Meta-Writing Ecology, this note functions as a Protocol-facing Note. It clarifies how LLM-mediated evidence requires condition-preserving documentation before it can be responsibly interpreted as research output.

It should be used when an LLM-based result is reported without enough information to reconstruct, audit, validate, or bound the execution environment that produced it.